

10 days
10 days
2 days
5 Days
3 days
2 days
5 days
10 days
5 days
10 days
5 days
10 days

Day 1 of 75 days Ethical Hacking Course

- Introduction
- Hacking labs setup
- Network fundamentals
- How websites works
- Linux fundamental
- Computer fundamentals
- Network security
-
- Wireless attacks
- System pentesting
- Social engineering
- Phishing
- Introduction to mobile hacking
- Some important tools in ethical hacking
- Wireshark, burpsuite, metasploit, autopsy, snort many more
- Introduction to webhacking
- Introduction to digital forensics
- Malware analysis
- Python basics
- Python for penetration testers
- Projects



- Introduction
- Hacking labs setup

Cyber security

Ethical hacking

offensive security

In short, offensive security is the process of breaking into computer systems, exploiting software bugs, and finding loopholes in applications to gain unauthorized access to them.

In a defensive cyber role, you could be investigating infected computers or devices to understand how it was hacked, tracking down cybercriminals, or monitoring infrastructure for malicious activity.

Which of the following options better represents the process where you simulate a hacker's actions to find vulnerabilities in a system?

- Offensive Security
- Defensive Security

.

HACK YOUR FIRST MACHINE

This page allows an attacker to steal money from any bank account, which is a critical risk for the bank. As an ethical hacker, you would (with permission) find vulnerabilities in their application and report them to the bank to fix before a hacker exploits them.

Transfer \$2000 from the bank account 2276, to your account (account number 8881)

<http://fakebank.com/bank-transfer> —

What careers are there?

The cyber careers room goes into more depth about the different careers in cyber. However, here is a short description of a few offensive security roles:

- Penetration Tester - Responsible for testing technology products for finding exploitable security vulnerabilities.
- Red Teamer - Plays the role of an adversary, attacking an organization and providing feedback from an enemy's perspective.
- Security Engineer - Design, monitor, and maintain security controls, networks, and systems to help prevent cyberattacks.

Defensive security



Defensive security is somewhat the opposite of offensive security, as it is concerned with two main tasks:

1. Preventing intrusions from occurring
2. Detecting intrusions when they occur and responding properly

Blue teams are part of the defensive security landscape.

Some of the tasks that are related to defensive security include:

- User cyber security awareness:
- Updating and patching systems:
- Setting up preventative security devices

Skills

- Security Operations Center (SOC)
- Threat Intelligence
- Digital Forensics and Incident Response (DFIR)
- Malware Analysis

Security Operations Center (SOC)

A *Security Operations Center (SOC)* is a team of cyber security professionals that monitors the network and its systems to detect malicious cyber security events. Some of the main areas of interest for a SOC are:

- Vulnerabilities:
- Unauthorized activity
- Network intrusions:
- .

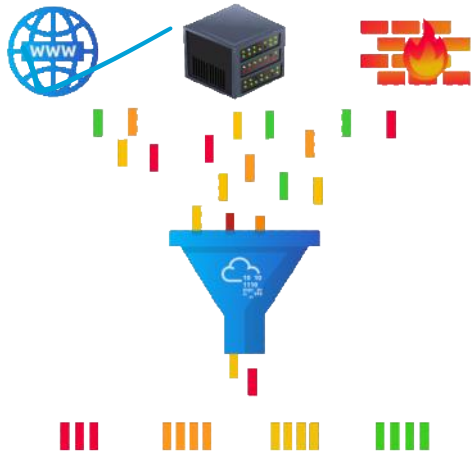
Security operations cover various tasks to ensure protection; one such task is threat intelligence.

Soc analyst



Threat Intelligence

In this context, *intelligence* refers to information you gather about actual and potential enemies. Threat intelligence aims to gather information to help the company better prepare against potential adversaries.



Intelligence needs data. Data has to be collected, processed, and analyzed. Data collection is done from local sources such as network logs and public sources such as forums. Processing of data aims to arrange them into a format suitable for analysis. The analysis phase seeks to find more information about the attackers and their motives; moreover, it aims to create a list of recommendations and actionable steps.

Learning about your adversaries allows you to know their tactics, techniques, and procedures. As a result of threat intelligence, we identify the threat actor (adversary), predict their activity, and consequently, we will be able to mitigate their attacks and prepare a response strategy.

Digital Forensics and Incident Response (DFIR)

This section is about Digital Forensics and Incident Response (DFIR), and we will cover:

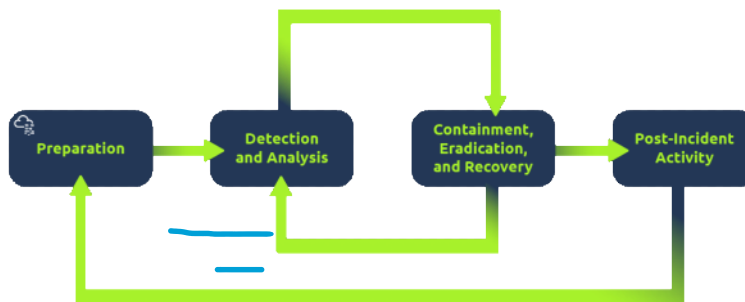
- Digital Forensics
- Incident Response
- Malware Analysis

Digital Forensics

Forensics is the application of science to investigate crimes and establish facts. With the use and spread of digital systems, such as computers and smartphones, a new branch of forensics was born to investigate related crimes: computer forensics, which later evolved into, *digital forensic*

Incident Response

An *incident* usually refers to a data breach or cyber attack; however, in some cases, it can be something less critical, such as a misconfiguration, an intrusion attempt, or a policy violation.



Malware Analysis

Malware stands for malicious software. *Software* refers to programs, documents, and files that you can save on a disk or send over the network. Malware includes many types, such as:

- Virus is a piece of code (part of a program) that attaches itself to a program. It is designed to spread from one computer to another; moreover, it works by altering, overwriting, and deleting files once it infects a computer. The result ranges from the computer becoming slow to unusable.
- Trojan Horse is a program that shows one desirable function but hides a malicious function underneath. For example, a victim might download a video player from a shady website that gives the attacker complete control over their system.
- Ransomware is a malicious program that encrypts the user's files. Encryption makes the files unreadable without knowing the encryption password. The attacker offers the user the encryption password if the user is willing to pay a "ransom."





https://ip-scanner.thm

IP-SCANNER.THM

Check by IP Address

Instructions

There are many open-source databases out there, like AbuseIPDB, and Cisco Talos Intelligence, where you can perform a reputation and location check for the IP address. Most security analysts use these tools to aid them with alert investigations. You can also make the Internet safer by reporting the malicious IPs, for example, on AbuseIPDB.

Now that we know the IP address is malicious, we need to escalate it to a staff member!

https://ip-scanner.thm/search

IP-SCANNER.THM

143.110.250.149 was found in our database!

Confidence of the IP being malicious is 100%

Malicious

ISP	China Mobile Communications Corporation
Domain Name	chinamobileltd.thm
Country	China
City	Zhenjiang, Jiangsu

Instructions

We shouldn't worry too much if it was a failed authentication attempt, but you probably noticed the successful authentication attempt from the malicious IP address. Let's declare a small incident event and escalate it. There is some great staff working at the company, but you wouldn't want to escalate this to the wrong person who is not in charge of your team or department.

Choose to whom you would escalate this event?

☐

Dominick Nash



Sales Executive

☐

Nadia Watson



Security Consultant

☐

Carolyn Stone



Information Security Architect

☒

Will Griffin



SOC Team Lead

Choose Staff Member

https://firewall.internal



Firewall Block List

Block List	
Date	IP Address
July 2nd 2021, 13:27:00:948	101.34.37.231
June 30th 2021, 09:12:11:857	212.38.99.12
June 23rd 2021, 23:56:28:370	213.106.84.35

143.110.250.149

Block IP Address

Day 2 of 75 days Ethical Hacking Course

lab setup
careers

Security analysts



Security analysts are important for keeping companies safe from cyber attacks. They look at how secure a company's computer systems are and suggest ways to make them safer. They work with different people in the company to understand what security is needed. Their job involves:

1. Checking how safe the company's computer networks are by working with different teams.
2. Writing reports regularly about what security problems they find and what they do to fix them.
3. Making plans to keep the company's data safe from new types of cyber attacks by doing research and working with different teams.

Security engineers



Security engineers develop and put in place security solutions using data on threats and vulnerabilities. They test security measures, monitor networks, and implement systems to keep data safe.

Incident responder

Incident responders react fast to security breaches, creating plans for organizations to follow during and after incidents. They work under pressure, responding to attacks in real-time. Key metrics include time to detect, acknowledge, and recover from attacks. The goal is to respond swiftly, protect the company's data, reputation, and finances, and learn from incidents for future readiness.

Digital Forensics



If you enjoy detective work, this job might be for you. Whether you're part of law enforcement or defending a company's network, you'll collect and analyze evidence to solve crimes or incidents. Responsibilities include:

- Collecting digital evidence following legal procedures
- Analyzing digital evidence to find answers
- Documenting findings and reporting on the case

Malware analysts





Handwritten notes: a bracket and the word "Powers" written diagonally.

Malware analysts investigate suspicious programs, uncovering their functions and writing reports. They reverse-engineer compiled programs to readable code, needing strong programming skills, especially in low-level languages like assembly and C. Their aim is to understand malicious program activities, develop detection methods, and report findings.

- Responsibilities:
- Perform static analysis by reverse-engineering
 - Conduct dynamic analysis in controlled environments
 - Document and report findings

Penetration testing,



Handwritten word: "Ethical"

Penetration testing, or pentesting, checks a company's systems for vulnerabilities through simulated hacking. Testers exploit these flaws to assess risks and recommend actions for prevention.

- Responsibilities:
- Test systems, networks, and web apps
 - Assess security and policies
 - Report insights and suggest preventive measures.

SALLERY

> SECURITY ANALYST

4L-8L

> Penetration testers

4L-10L

> Malware analyst

4L-9L

> Incident responder

4L-15L

> Digital forensics

5L-7L

> SECURITY ENGINEER

6L-16L

Handwritten notes: "Skill" and "LOC" with a bracket underneath.

Handwritten text: "4LA - 8LA"

Handwritten notes: "4L 5L" and a circled "35K".

Handwritten text: "4L - 15L" inside a large bracket.

4L-15L

S-
8G-16L

92V ✓

✓ ✓ ✓

✓ (CBI) →

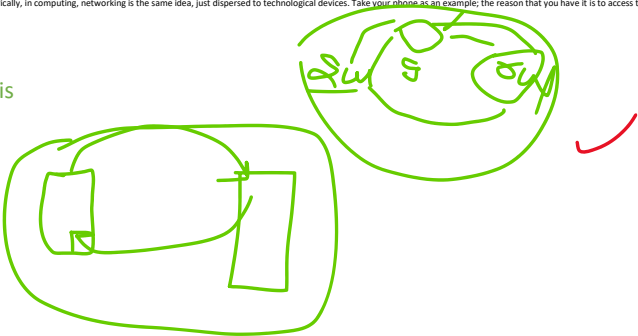
What is Networking?

Networks are simply things connected. For example, your friendship circle: you are all connected because of similar interests, hobbies, skills and sorts. Networks can be found in all walks of life:

- A city's public transportation system
- Infrastructure such as the national power grid for electricity
- Meeting and greeting your neighbours
- Postal systems for sending letters and parcels

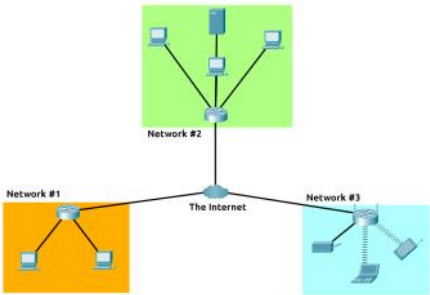
But more specifically, in computing, networking is the same idea, just dispersed to technological devices. Take your phone as an example: the reason that you have it is to access things. We'll cover how these devices communicate with each other and the rules that follow.

What is



the Internet?

Now that we've learnt what a network is and how one is defined in computing (just devices connected), let's explore the Internet. The Internet is one giant network that consists of many, many small networks within itself. Using our example from the previous task,



As previously stated, the Internet is made up of many small networks all joined together. These small networks are called private networks, where networks connecting these small networks are called public networks -- or the Internet! So, to recap, a network can be one of two types:

- A private network
- A public network

Identifying Devices on a Network

Devices on a network are very similar to humans in the fact that we have two ways of being identified:

- Our Name
- Our Fingerprints

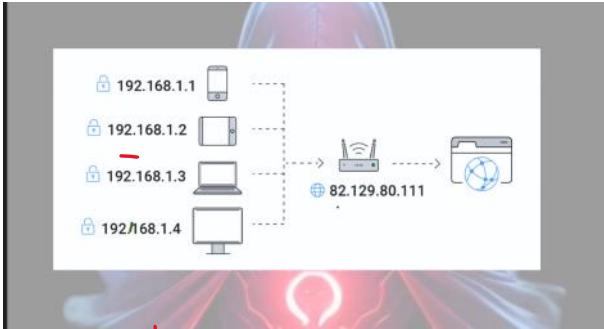
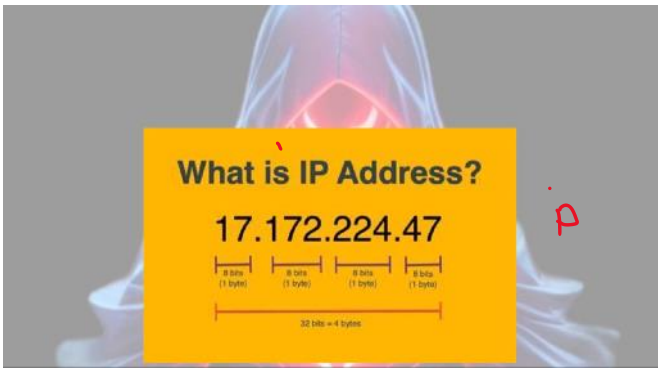
Now we can change our name through deed poll, but we can't, however, change our fingerprints. Every human has an individual set of fingerprints which means that even if they change their name, there is still an identity behind it. Devices have the same thing: two means of identification, with one being permeable. These are:

- An IP Address
- A Media Access Control (MAC) Address -- think of this as being similar to a serial number.



IP Addresses

Briefly, an IP address (or Internet Protocol) address can be used as a way of identifying a host on a network for a period of time, where that IP address can then be associated with another device without the IP address changing. First, let's split up precisely what an IP address is in the diagram below:

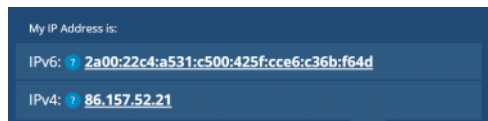


Cisco, an industry giant in the world of networking, estimated that there would be approximately 50 billion devices connected on the Internet by the end of 2021. [Cisco, 2021](#). Enter IP address versions. So far, we have only discussed one version of the Internet Protocol addressing scheme known as IPv4, which uses a numbering system of 2^{32} IP addresses (4.29 billion) – so you can see why there is such a shortage!

IPv6 is a new iteration of the Internet Protocol addressing scheme to help tackle this issue. Although it is seemingly more daunting, it boasts a few benefits:

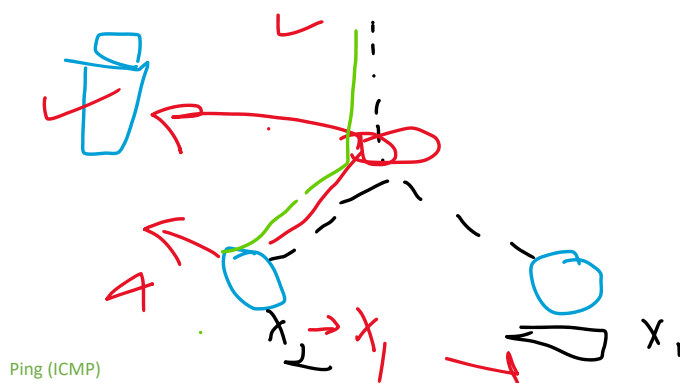
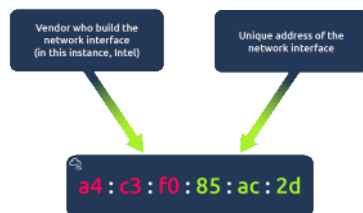
- Supports up to 2^{128} of IP addresses (340 trillion-plus), resolving the issues faced with IPv4
- More efficient due to new methodologies

The screenshot below compares both an IPv6 and IPv4 address.



MAC Addresses

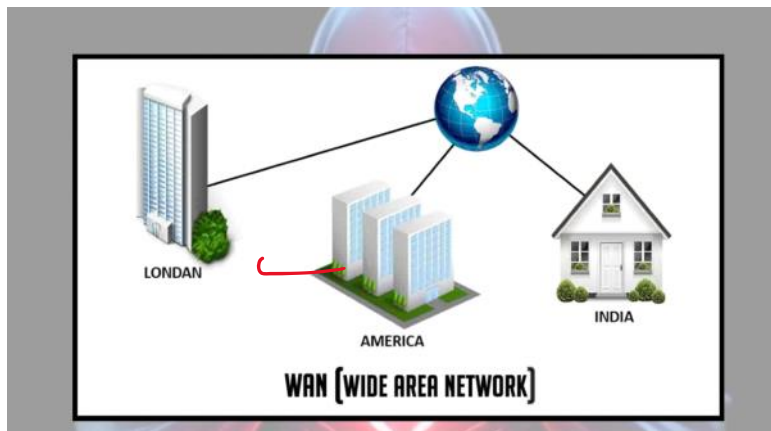
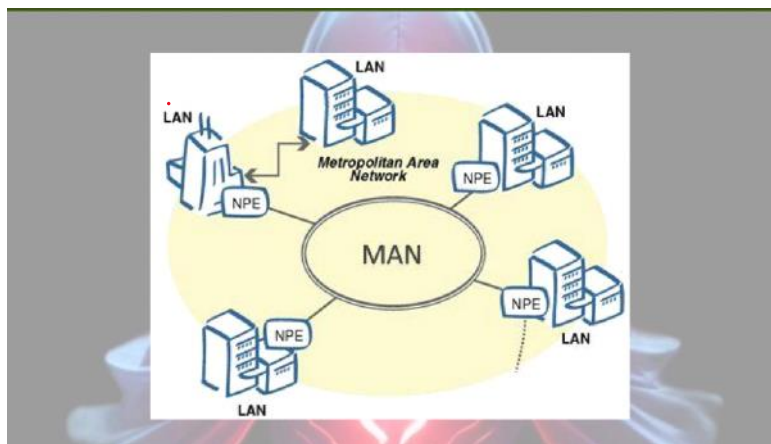
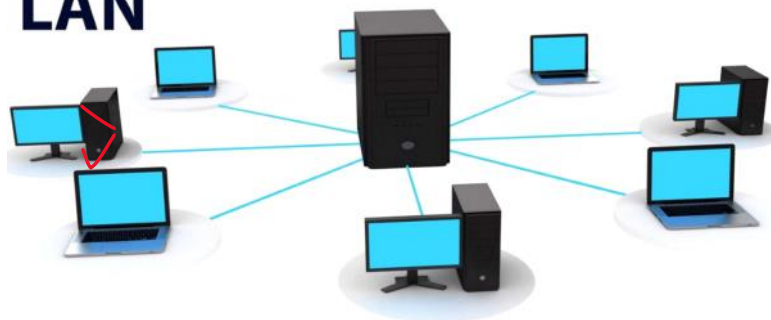
Devices on a network will all have a physical network interface, which is a microchip found on the device's motherboard. This network interface is assigned a unique address at the factory it was built at, called a **MAC** (Media Access Control) address. `04:c3:f0:85:ac:2d`. The first six characters represent the company that made the network interface, and the last six is a unique number.



Ping is one of the most fundamental network tools available to us. Ping uses **ICMP** (Internet Control Message Protocol) packets to determine the performance of a connection between devices, for example, if the connection exists or is reliable. The time taken for ICMP packets travelling between devices is measured by ping, such as in the screenshot below. This measuring is done using ICMP's echo packet and then ICMP's echo reply from the target device.

Type of networks

LAN



Introducing LAN Topologies

Local Area Network (LAN) Topologies

Over the years, there has been experimentation and implementation of various network designs. In reference to networking, when we refer to the term "topology", we are actually referring to the design or look of the network at hand. Let's discuss the advantages and disadvantages of these topologies below.

Star Topology

The main premise of a star topology is that devices are individually connected via a central networking device such as a switch or hub. This topology is the most commonly found today because of its reliability and scalability—despite the cost.

Bus Topology

This type of connection relies upon a single connection which is known as a backbone cable. This type of topology is similar to the leaf off of a tree in the sense that devices (leaves) stem from where the branches are on this cable.

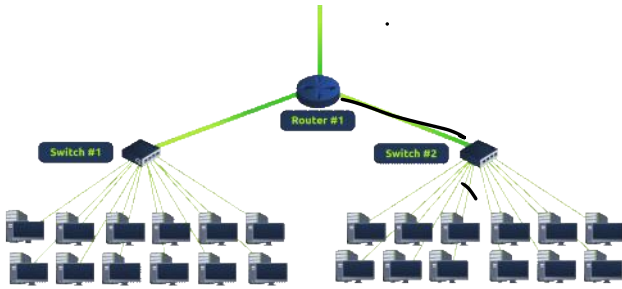
Ring Topology

The ring topology (also known as token topology) boasts some similarities. Devices such as computers are connected directly to each other to form a loop, meaning that there is little cabling required and less dependence on dedicated hardware such as within a star topology.

What is a Switch?

Switches are dedicated devices within a network that are designed to aggregate multiple other devices such as computers, printers, or any other networking-capable device using ethernet.

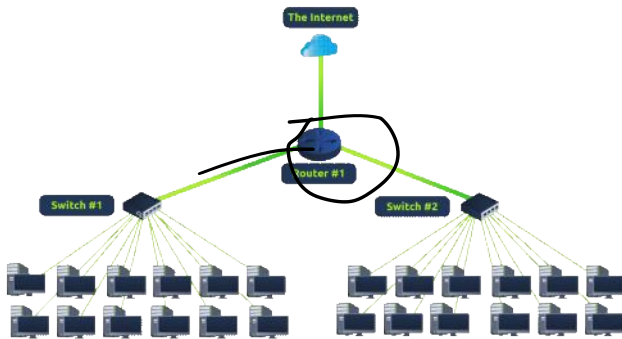




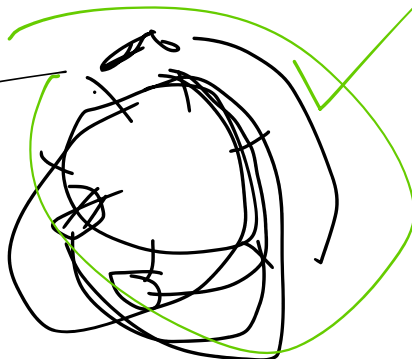
Both Switches and Routers can be connected to one another. The ability to do this increases the redundancy (the reliability) of a network by adding multiple paths for data to take. If one path goes down, another can be used. Whilst this may reduce the overall performance of a network because packets have to take longer to travel, there is no downtime – a small price to pay considering the alternative.

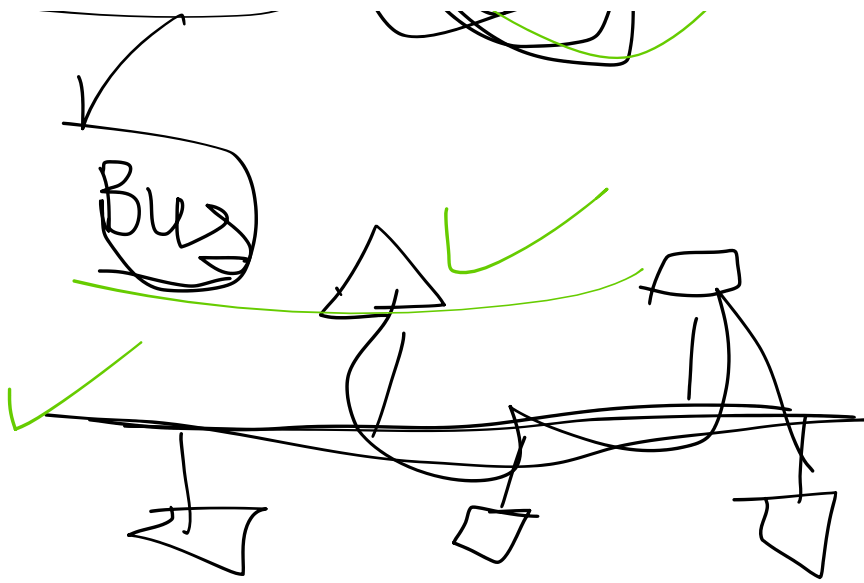
What is a Router?

It's a router's job to connect networks and pass data between them. It does this by using routing (hence the name router!). Routing is the label given to the process of data travelling across networks. Routing involves creating a path between networks so that this data can be successfully delivered.

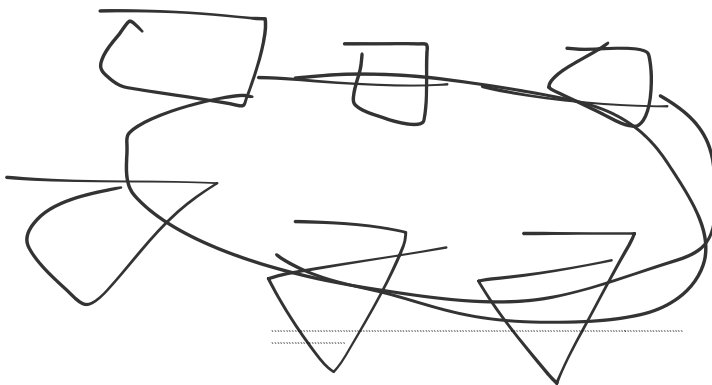
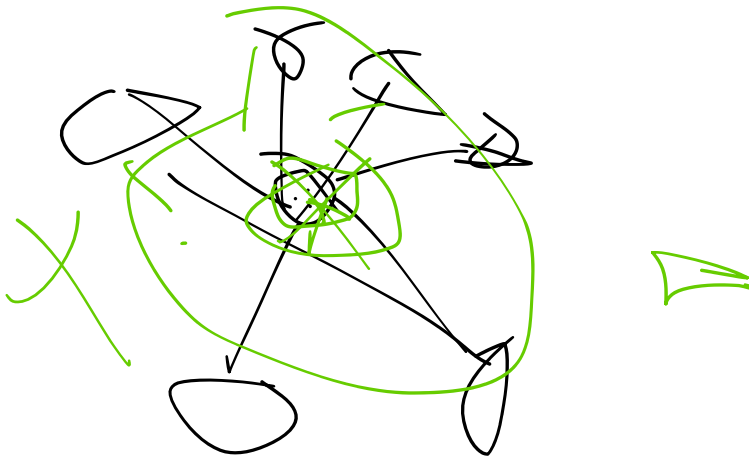


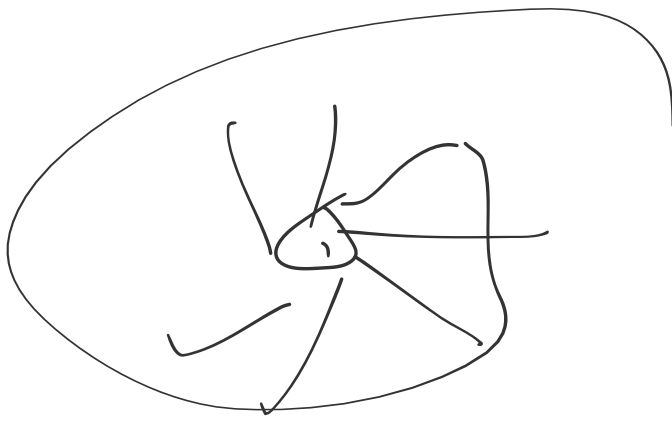
Ring ✓





of the ✓





DAY 4 OF 75 DAYS ETHICAL H4CKING COURSE

- Introduction
- Hacking labs setup
- Network fundamentals
- How websites works
- Linux fundamental
- Computer fundamentals
- Network security
- Wireless attacks
- System pentesting
- Social engineering
- Phishing
- Introduction to mobile hacking
- Some important tools in ethical hacking
- Wireshark, burpsuite, metasploit, autopsy, snort many more
- Introduction to webhacking
- Introduction to digital forensics
- Malware analysis
- Python basics
- Python for penetration testers
- Projects

- Introduction
- Hacking labs setup
- Network fundamentals
- websites working and technology
- Linux fundamental
- COMPUTER FUNDA

OSI MODEL
PACKETS
PORT FORWARDING
VPN
FIREWALLS ETC

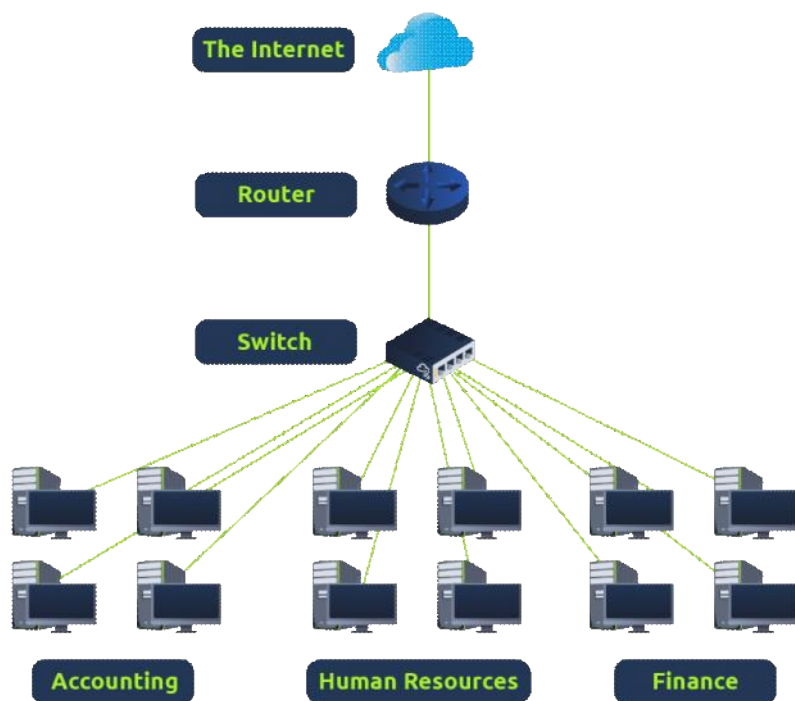
Subnetting

As we've previously discussed throughout the module so far, Networks can be found in all shapes and sizes - ranging from small to large. Subnetting is the term given to splitting up a network into smaller, miniature networks within itself.

Think of it as slicing up a cake for your friends. There's only a certain amount of cake to go around, but everybody wants a piece. Subnetting is you deciding who gets what slice & reserving such a slice of this metaphorical cake.

Take a business, for example; You will have different departments such as:

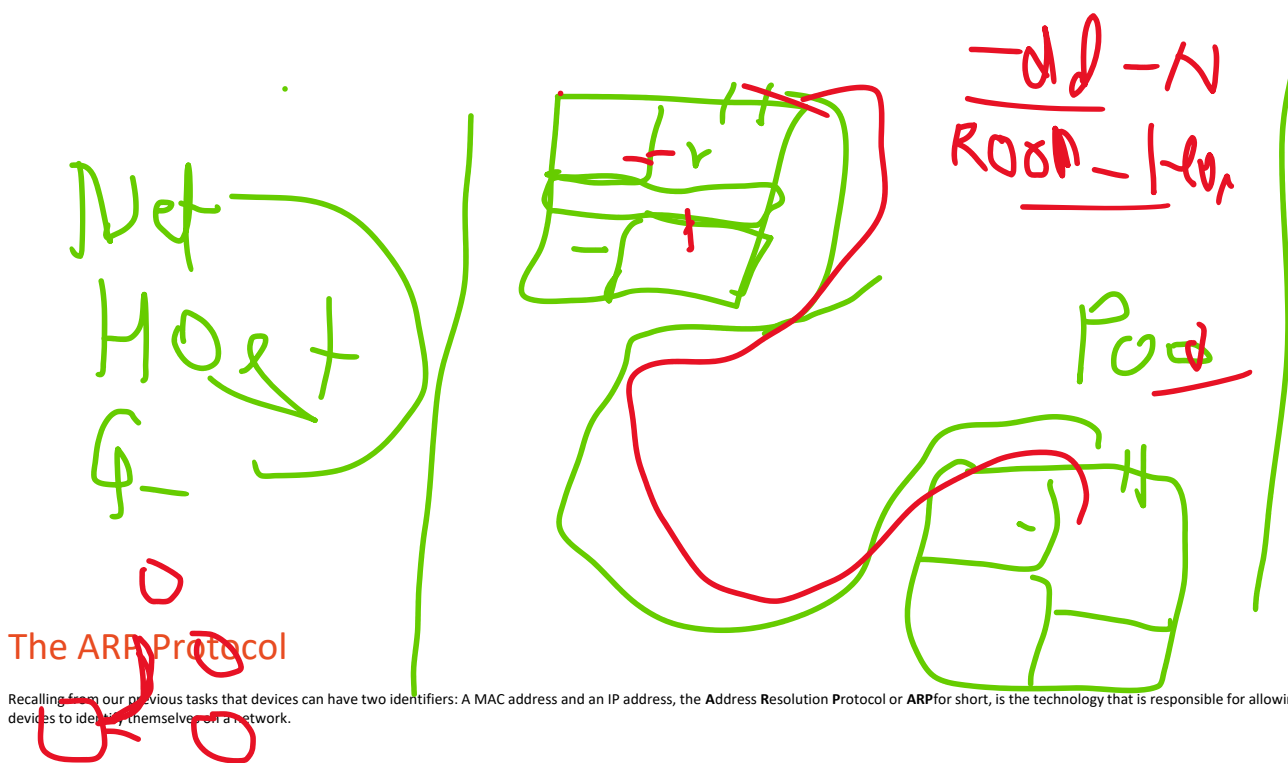
- Accounting
- Finance



Whilst you know where to send information in real life to the correct department, networks need to know as well. Network administrators use subnetting to categorise and assign specific parts of a network to reflect this.
Subnetting is achieved by splitting up the number of hosts that can fit within the network, represented by a number called a subnet mask. Let's refer back to our diagram from the first room in this module:

As we can recall, an IP address is made up of four sections called octets. The same goes for a subnet mask which is also represented as a number of four bytes (32 bits), ranging from 0 to 255 (0-255). Subnets use IP addresses in three different ways:

- Identify the network address
- Identify the host address
- Identify the default gateway



Recalling from our previous tasks that devices can have two identifiers: A MAC address and an IP address, the Address Resolution Protocol or ARP for short, is the technology that is responsible for allowing devices to identify themselves on a network.

Recalling from our previous tasks that devices can have two identifiers: A MAC address and an IP address, the **Address Resolution Protocol** or **ARP** for short, is the technology that is responsible for allowing devices to identify themselves on a network.

Simply, the ARP protocol allows a device to associate its MAC address with an IP address on the network. Each device on a network will keep a log of the MAC addresses associated with other devices.

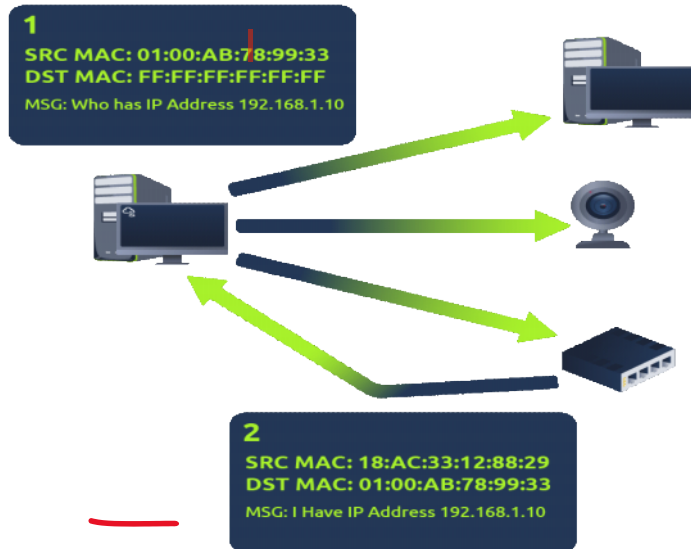
When devices wish to communicate with another, they will send a broadcast to the entire network searching for the specific device. Devices can use the ARP protocol to find the MAC address (and therefore the physical identifier) of a device for communication.

How does ARP Work?

Each device within a network has a ledger to store information on, which is called a cache. In the context of the **ARP** protocol, this cache stores the identifiers of other devices on the network. In order to map these two identifiers together (IP address and MAC address), the ARP protocol sends two types of messages:

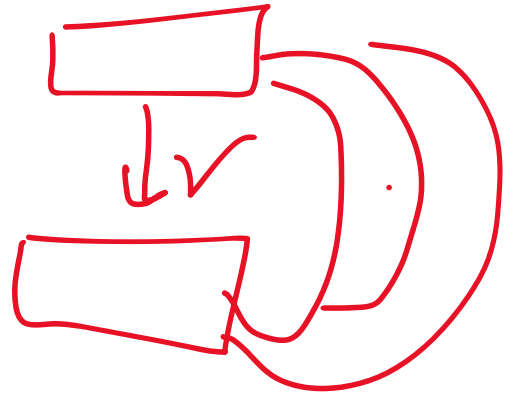
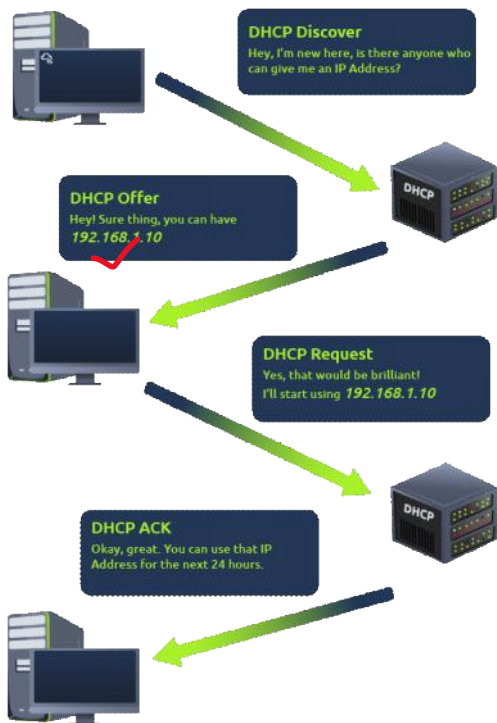
1. **ARPRequest**
2. **ARPReply**

When an **ARPRequest** is sent, a message is broadcasted to every other device found on a network by the device, asking whether or not the device's MAC address matches the requested IP address. If the device does have the requested IP address, an **ARPReply** is returned to the initial device to acknowledge this. The initial device will now remember this and store it within its cache (an ARPentry). This process is illustrated in the diagram below:



The DHCP Protocol

IP addresses can be assigned either manually, by entering them physically into a device, or automatically and most commonly by using a **DHCP**(Dynamic Host Configuration Protocol) server. When a device connects to a network, if it has not already been manually assigned an IP address, it sends out a request (DHCPDiscover) to see if any DHCP servers are on the network. The DHCP server then replies back with an IP address the device could use (DHCPOffer). The device then sends a reply confirming it wants the offered IP Address (DHCPRequest), and then lastly, the DHCP server sends a reply acknowledging this has been completed, and the device can start using the IP Address (DHCPACK).



QUIES

What is the technical term for dividing a network up into smaller pieces?

subnetting

How many **bits** are in a subnet mask?

32

What is the range of a section (octet) of a subnet mask?

0-255

What address is used to identify the start of a network?

network address

What address is used to identify devices within a network?

host address

What does ARP stand for?

address resolution protocol

What category of ARP Packet asks a device whether or not it has a specific IP address?

request

What address is used as a physical identifier for a device on a network?

mac addresss

What address is used as a logical identifier for a device on a network?

ip address

What type of DHCP packet is used by a device to **retrieve an IP address**?

dhcp discover

What type of DHCP packet does a device **send once it has been offered an IP address** by the DHCP server?

dhcp request

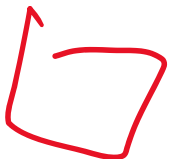
Finally, **what is the last** DHCP packet that is sent to a device from a DHCP server?

dhcp ack

255.255.255.0
- 172.20.10.10 ✓

172.20.10.0

172.20.10.1, 2, 3, 4
- - - 254 ✓

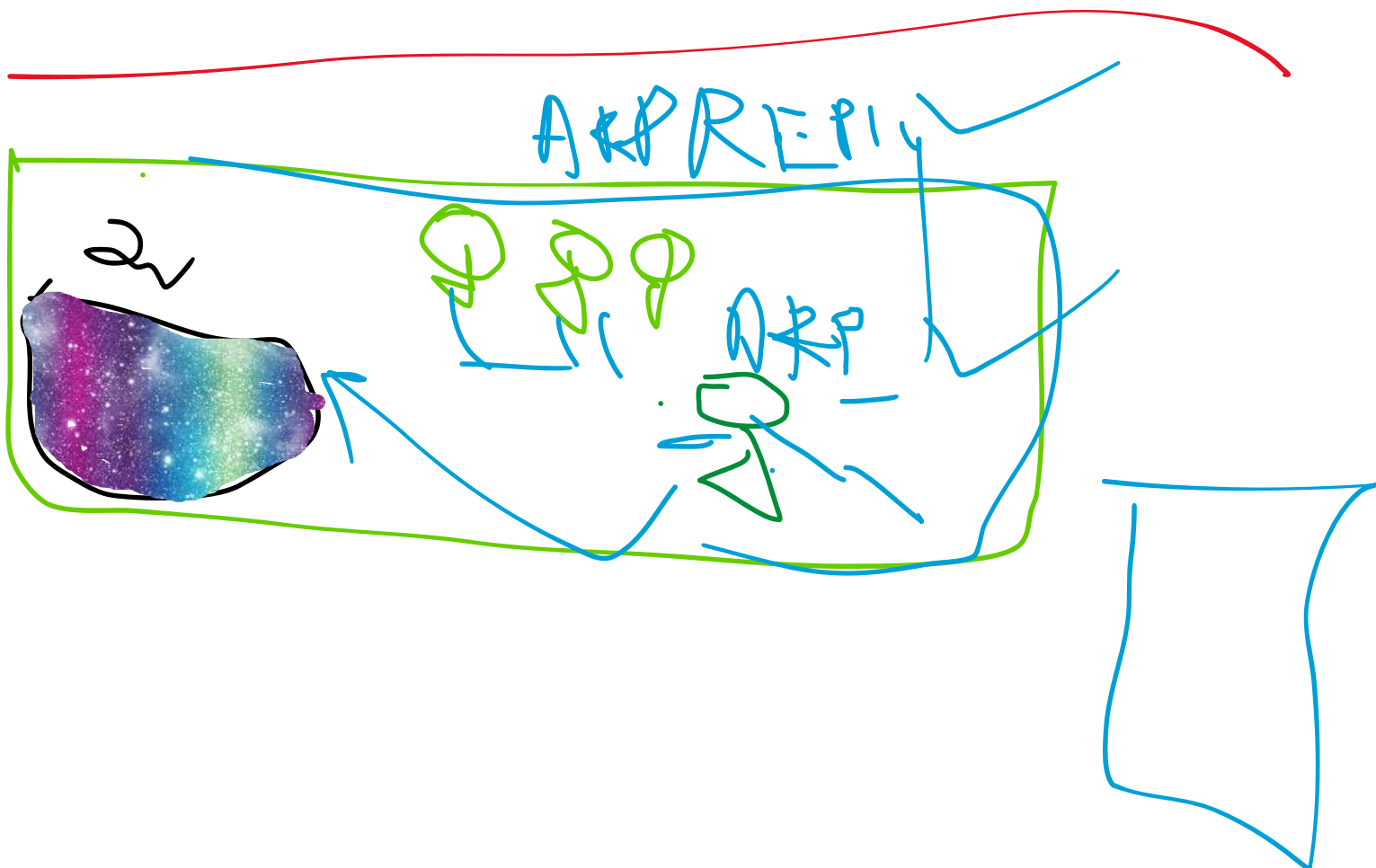


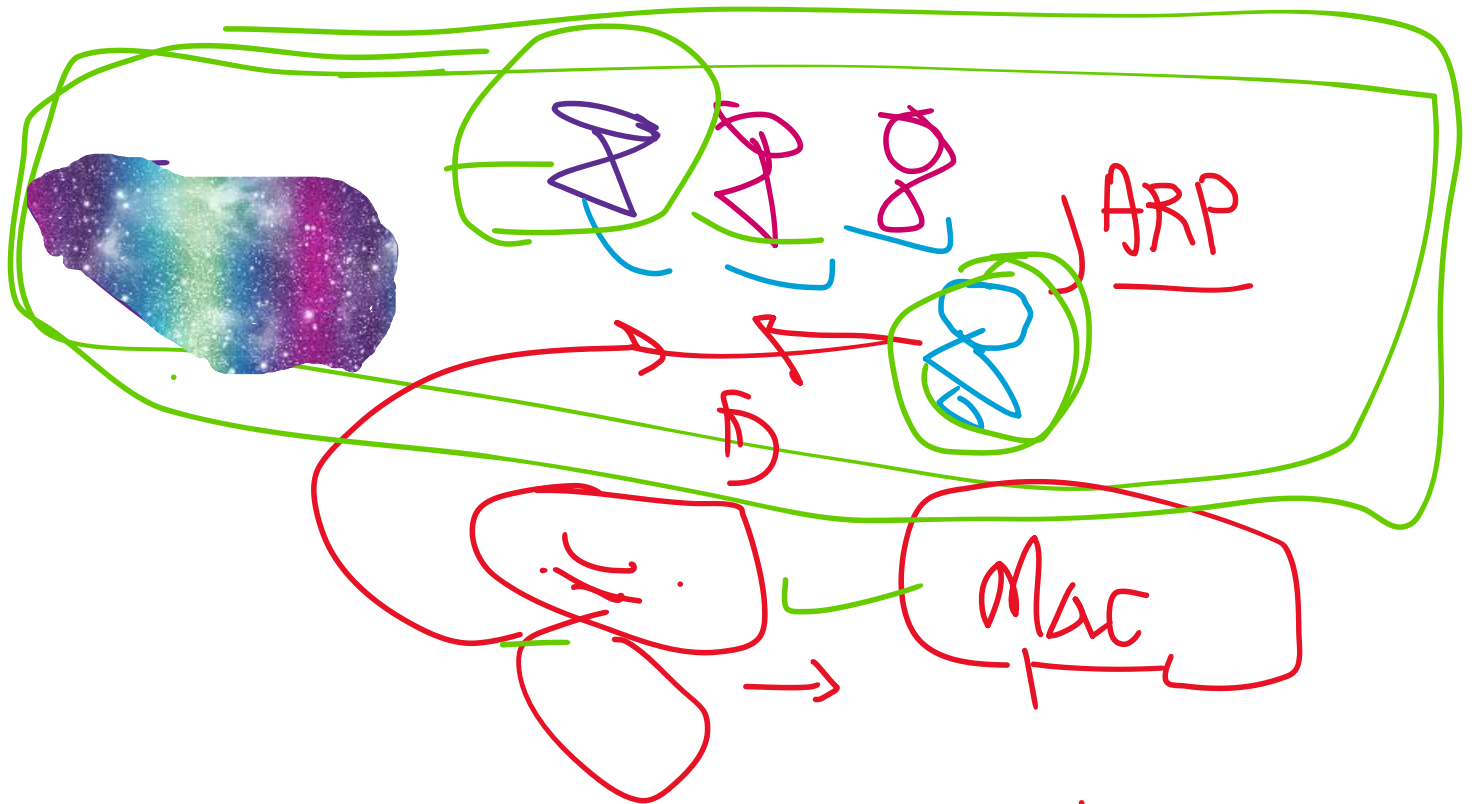
172.20.10.255

172.20.10.0 network
addr

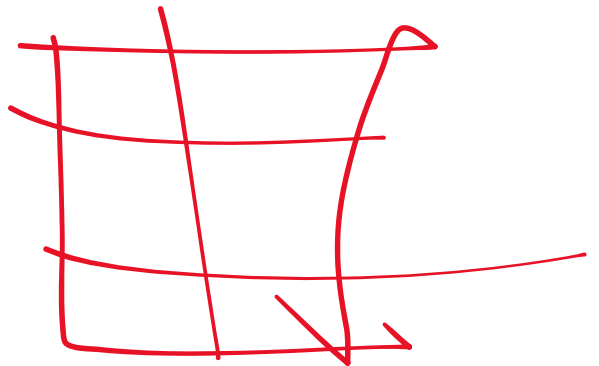
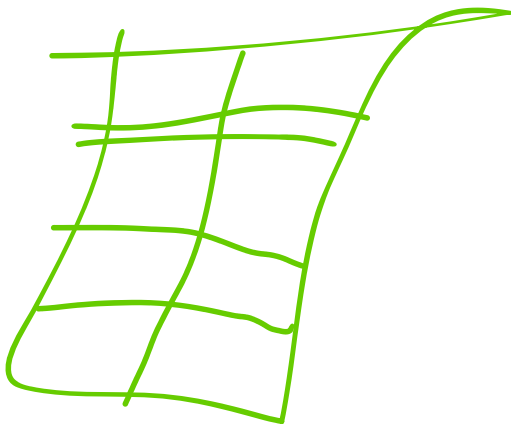
172.20.10.1-254 - host
addr

172.20.10.255-
gateway





ON

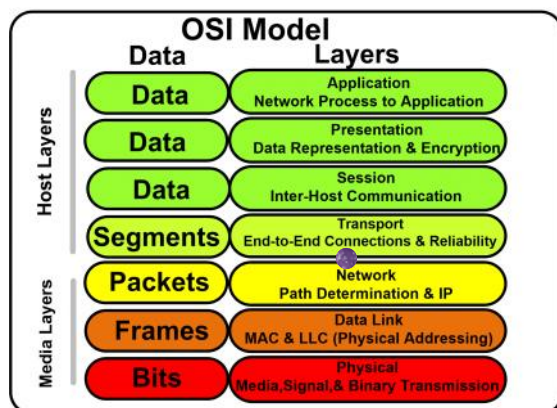


DAY 5 OF 75 DAYS ETHICAL HACKING COURSE

The **OSI** model (or **Open Systems Interconnection Model**) is an absolute fundamental model used in networking. This critical model provides a framework dictating how all networked devices will send, receive and interpret data.

One of the main benefits of the OSI model is that devices can have different functions and designs on a network while communicating with other devices. Data sent across a network that follows the uniformity of the OSI model can be understood by other devices.

The OSI model consists of seven layers which are illustrated in the diagram below. Each layer has a different set of responsibilities and is arranged from Layer 7 to Layer 1. At every individual layer that data travels through, specific processes take place, and pieces of information are added to this data, which is what we'll come to discuss in the upcoming tasks within this room. However, for now, we only need to understand that this process is called encapsulation and what the OSI model looks like in the diagram below:



M

APPLICATION

The application layer of the OSI model is the layer that you will be most familiar with. This familiarity is because the application layer is the layer in which protocols and rules are in place to determine how the user should interact with data sent or received.

Everyday applications such as email clients, browsers, or file server browsing software such as FileZilla provide a friendly, Graphical User Interface (GUI) for users to interact with data sent or received. Other protocols include **DNS** (Domain Name System), which is how website addresses are translated into IP addresses.



What is the technical term that is given to the name of the software that users interact with?

graphical user interface

Layer 6 - Presentation

Layer 6 of the OSI model is the layer in which standardisation starts to take place. Because software developers can develop any software such as an email client differently, the data still needs to be handled in the same way — no matter how the software works.

This layer acts as a translator for data to and from the application layer (layer 7). The receiving computer will also understand data sent to a computer in one format destined for in another format. For example, when you send an email, the other user may have another email client to you, but the contents of the email will still need to display the same.

Security features such as data encryption (like HTTPS when visiting a secure site) occur at this layer.

What is the main purpose that this Layer acts as?

translator

Layer 5 - Session

Once data has been correctly translated or formatted from the presentation layer (layer 6), the session layer (layer 5) will begin to create a connection to the other computer that the data is destined for. When a connection is established, a session is created. Whilst this connection is active, so is the session.

The session layer (layer 5) synchronises the two computers to ensure that they are on the same page before data is sent and received. Once these checks are in place, the session layer will begin to divide up the data sent into smaller chunks of data and begin to send these chunks (packets) one at a time. This dividing up is beneficial because if the connection is lost, only the chunks that weren't yet sent will have to be sent again — not the entire piece of the data (think of it as loading a save file in a video game).

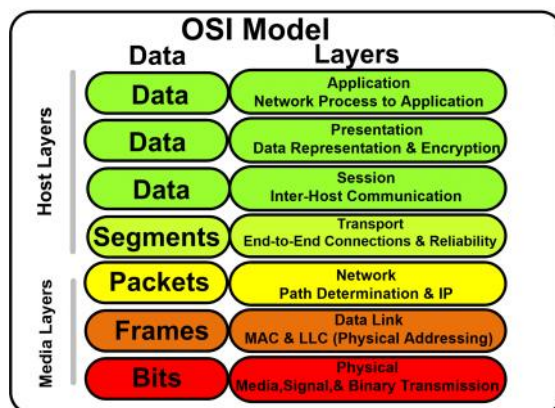
What is worthy of noting is that sessions are unique — meaning that data cannot travel over different sessions, but in fact, only across each session instead



What is the technical term for "small chunks of data"?

packets

Layer 4 - Transport



Layer 4 of the OSI model plays a vital part in transmitting data across a network and can be a little bit difficult to grasp. When data is sent between devices, it follows one of two different protocols that are decided based upon several factors:

- TCP
- UDP

Let's begin with TCP. The **T**ransmission **C**ontrol **P**rotocol (**TCP**). Potentially hinted by the name, this protocol is designed with reliability and guarantee in mind. This protocol

reserves a constant connection between the two devices for the amount of time it takes for the data to be sent and received.
Not only this, but TCP incorporates error checking into its design. Error checking is how TCP can guarantee that data sent from the small chunks in the session layer (layer 5) has then been received and reassembled in the same order.

Let's summarise the advantages and disadvantages of TCP in the table below:

Advantages of TCP

Guarantees the accuracy of data.

Capable of synchronising two devices to prevent each other from being flooded with data.

Performs a lot more processes for reliability.

Disadvantages of TCP

Requires a reliable connection between the two devices. If one small chunk of data is not received, then the entire chunk of data cannot be used

TCP is significantly slower than UDP because more work has to be done by the devices using this protocol.

TCP is used for situations such as file sharing, internet browsing or sending an email. This usage is because these services require the data to be accurate and complete (no good having half a file!).

Now let's move onto the **User Datagram Protocol** (or **UDP** for short). This protocol is not nearly as advanced as its brother - the TCP protocol. It doesn't boast the many features offered by TCP, such as error checking and reliability. In fact, any data that gets sent via UDP is sent to the computer whether it gets there or not. There is no synchronisation between the two devices or guarantee; just hope for the best, and fingers crossed.

Whilst this sounds disadvantageous, it does have its merits, which we'll layout in the table below:

Advantages of UDP

UDP is much faster than TCP.

UDP leaves the application layer (user software) to decide if there is any control over how quickly packets are sent.

UDP does not reserve a continuous connection on a device as TCP does.

Disadvantages of UDP

UDP doesn't care if the data is received.

It is quite flexible to software developers in this sense.

This means that unstable connections result in a terrible experience for the user.

Using the same example as before, we can now see that only Packets #1 and #3 have been received by the "Computer", meaning that half of the image is missing. UDP is useful in situations where there are small pieces of data being sent. For example, protocols used for discovering devices (*ARP* and *DHCP*) that we discussed

Answer the questions below

What is the name of this Layer?

transport

What does TCP stand for?

transmission control protocol

What does UDP stand for?

user datagram protocol

What protocol guarantees the accuracy of data?

tcp

What protocol doesn't care if data is received or not by the other device?

udp

Layer 3 - Network

The third layer of the OSI model (network layer) is where the magic of routing & re-assembly of data takes place (from these small chunks to the larger chunk). Firstly, routing simply determines the most optimal path in which these chunks of data should be sent. Whilst some protocols at this layer determine exactly what is the "optimal" path that data should take to reach a device, we should only know about their existence at this stage of the networking module. Briefly, these protocols include **OSPF (Open Shortest Path First)** and **RIP (Routing Information Protocol)**. The factors that decide what route is taken is decided by the following:

- What path is the shortest? I.e. has the least amount of devices that the packet needs to travel across.
- What path is the most reliable? I.e. have packets been lost on that path before?
- Which path has the faster physical connection? I.e. is one path using a copper connection (slower) or a fibre (considerably faster)?

At this layer, everything is dealt with via IP addresses such as 192.168.1.100. Devices such as routers capable of delivering packets using IP addresses are known as Layer 3 devices — because they are capable of working at the third layer of the OSI model.

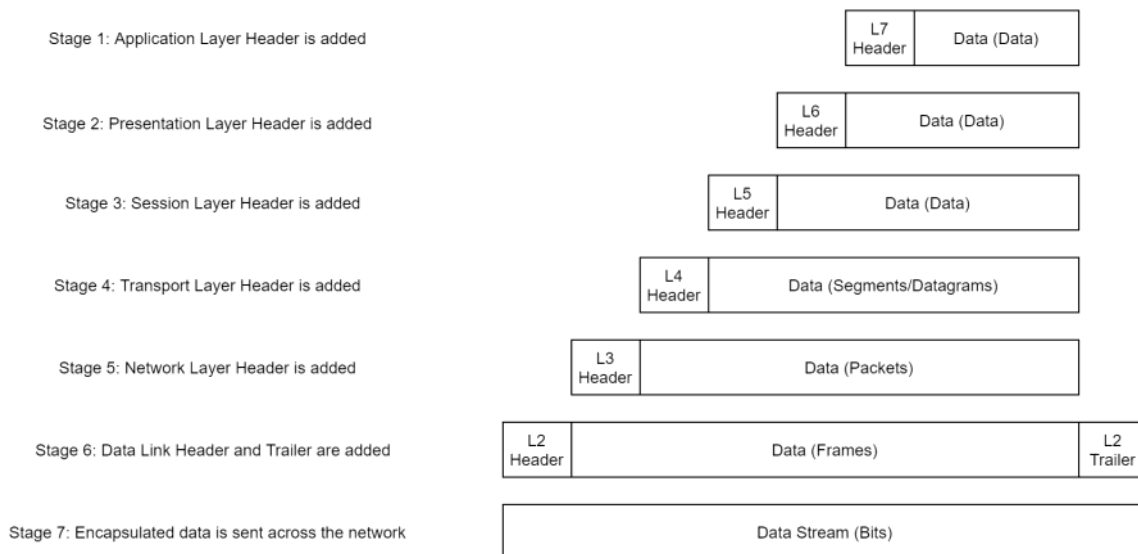
Layer 2 - Data Link

The data link layer focuses on the physical addressing of the transmission. It receives a packet from the network layer (including the IP address for the remote computer) and adds in the physical **MAC (Media Access Control)** address of the receiving endpoint. Inside every network-enabled computer is a **Network Interface Card (NIC)** which comes with a unique MAC address to identify it.

MAC addresses are set by the manufacturer and literally burnt into the card; they can't be changed -- although they can be spoofed. When information is sent across a network, it's actually the physical address that is used to identify where exactly to send the information. Additionally, it's also the job of the data link layer to present the data in a format suitable for transmission.

Layer 1 - Physical

This layer is one of the easiest layers to grasp. Put simply, this layer references the physical components of the hardware used in networking and is the lowest layer that you will find. Devices use electrical signals to transfer data between each other in a binary numbering system (1's and 0's).



TCP/IP MODEL

The TCP/IP model is, in many ways, very similar to the OSI model. It's a few years older, and serves as the basis for real-world networking. The TCP/IP model consists of four layers: Application, Transport, Internet and Network Interface. Between them, these cover the same range of functions as the seven layers of the OSI Model.

TCP/IP
Application
Transport
Internet
Network Interface

Note: Some recent sources split the TCP/IP model into five layers -- breaking the Network Interface layer into Data Link and Physical layers (as with the OSI model). This is accepted and well-known; however, it is not officially defined (unlike the original four layers which are defined in RFC1122). It's up to you which version you use -- both are generally considered valid.

You would be justified in asking why we bother with the OSI model if it's not actually used for anything in the real-world. The answer to that question is quite simply that the OSI model (due to being less condensed and more rigid than the TCP/IP model) tends to be easier for learning the initial theory of networking. The two models match up something like this:

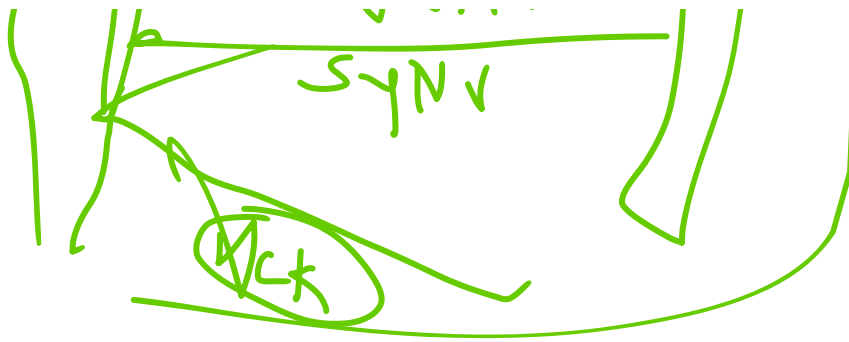
OSI	TCP/IP
Application	Application
Presentation	
Session	
Transport	Transport
Network	Internet
Data Link	Network Interface
Physical	

• TCP Flags:

Flag	Name	Function
SYN	Synchronize	Set during initial communication. Negotiating of parameters and sequence numbers (SYN(Synchronised) - establish the connection between hosts and devices)
ACK	Acknowledgment	Set as an acknowledgement to the SYN flag. Always set after initial SYN(ACK(Acknowledgement) - means this verify if the data is reached there or not)
RST	Reset	Forces the termination of a connection (in both directions)(RST(Reset) - resets a connection)
FIN	Finish	Ordered close to communications(FIN(finish) - No further transmission)
PSH	Push	Forces the delivery of data without concern for buffering(PSH(PUSH) - Send the buffer data immediately so the system can finish connection)
URG	Urgent	Data inside is being sent out of band. Example is cancelling a message(URG(urgent) - data contained in the packet should be processed immediately)

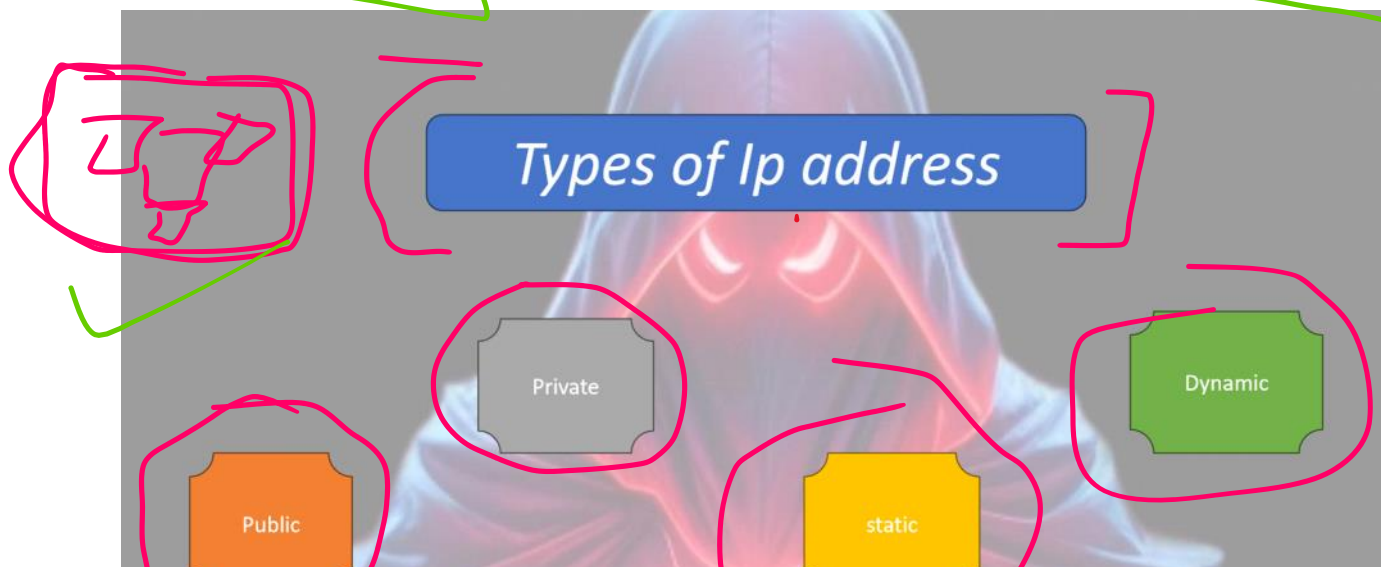
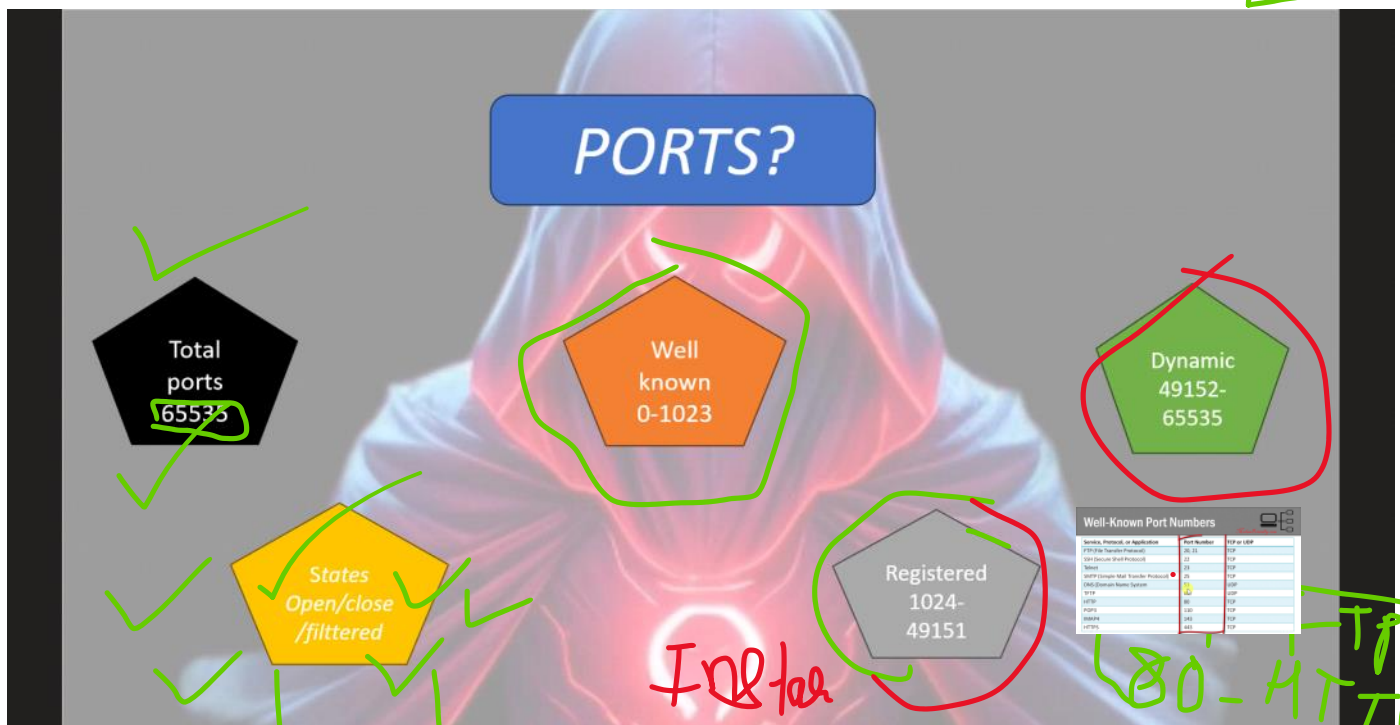
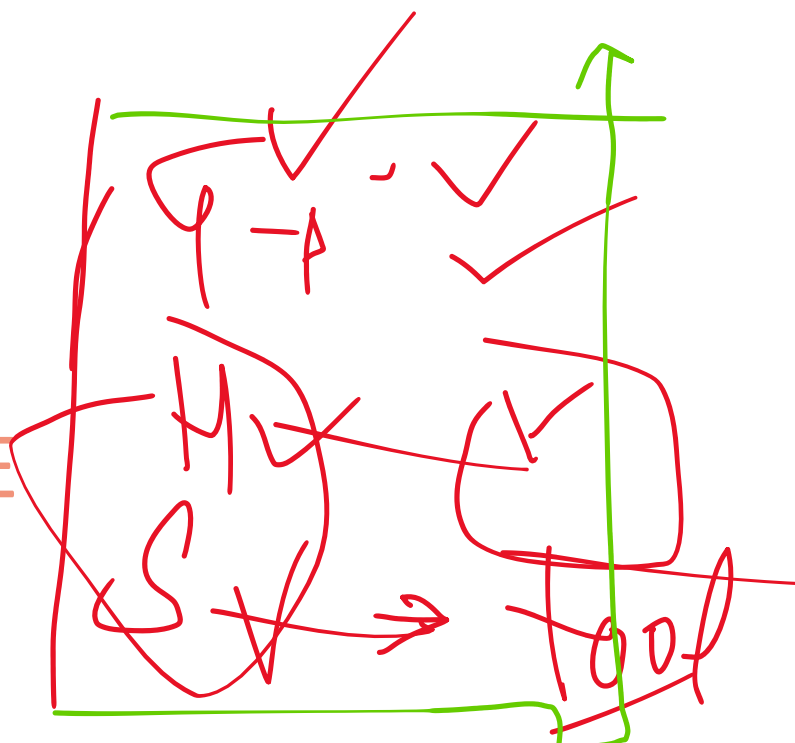
History:

It's important to understand exactly why the TCP/IP and OSI models were originally created. To begin with there was no standardisation -- different manufacturers followed their own methodologies, and consequently systems made by different manufacturers were completely incompatible when it came to networking. The TCP/IP model was introduced by the American DoD in 1982 to provide a standard -- something for all of the different manufacturers to follow. This sorted out the inconsistency problems. Later the OSI model was also introduced by the International Organisation for Standardisation (ISO); however, it's mainly used as a more comprehensive guide for learning, as the TCP/IP model is still the standard upon which modern networking is based.



DAY 6 OF 75 DAY ETHICAL HACKING COURSE

Ports





~~Static~~

FIREWALLS

PORT FORWARDING

DATA PACKETS AND FRAMES

THREE WAY HANDSHAKE

Flag	Name	Function
SYN	Synchronize	Set during initial communication. Negotiating of parameters and sequence numbers (SYN(Synchronised) - establish the connection between hosts and devices)
ACK	Acknowledgment	Set as an acknowledgement to the SYN flag. Always set after initial SYN(ACK(Acknowledgement) - means this verify if the data is reached there or not)
RST	Reset	Forces the termination of a connection (in both directions)(RST(Reset) - resets a connection)
FIN	Finish	Ordered close to communications(FIN(finish) - No further transmission)
PSH	Push	Forces the delivery of data without concern for buffering(PSH(PUSH) - Send the buffer data immediately so the system can finish connection)
URG	Urgent	Data inside is being sent out of band. Example is cancelling a message(Urg(urgent) - data contained in the packet should be processed indiatly)

The TCP Three-way handshake:

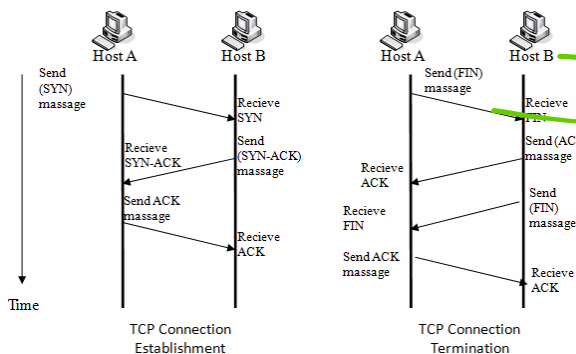


Figure 2.1. TCP session establishment and termination

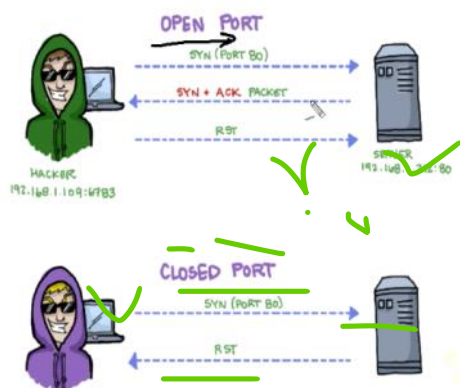
Handwritten notes in green ink:

- A large green checkmark.
- Handwritten "SYN" with a superscript "2" (SYN²).
- Handwritten "ACK + SYN" with a checkmark.

Ack + SYN
21
ACK

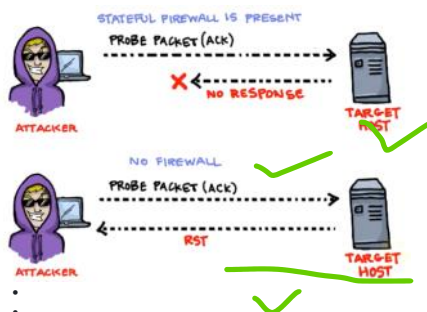
Port Discovery - Basic Concepts

Knocking the door:



- The hacker above sends a SYN packet to port 80 on the server.
 - If server returns SYN-ACK packet = the port is open
 - If server returns RST (reset) packet = the port is closed

Checking if Stateful Firewall is present:

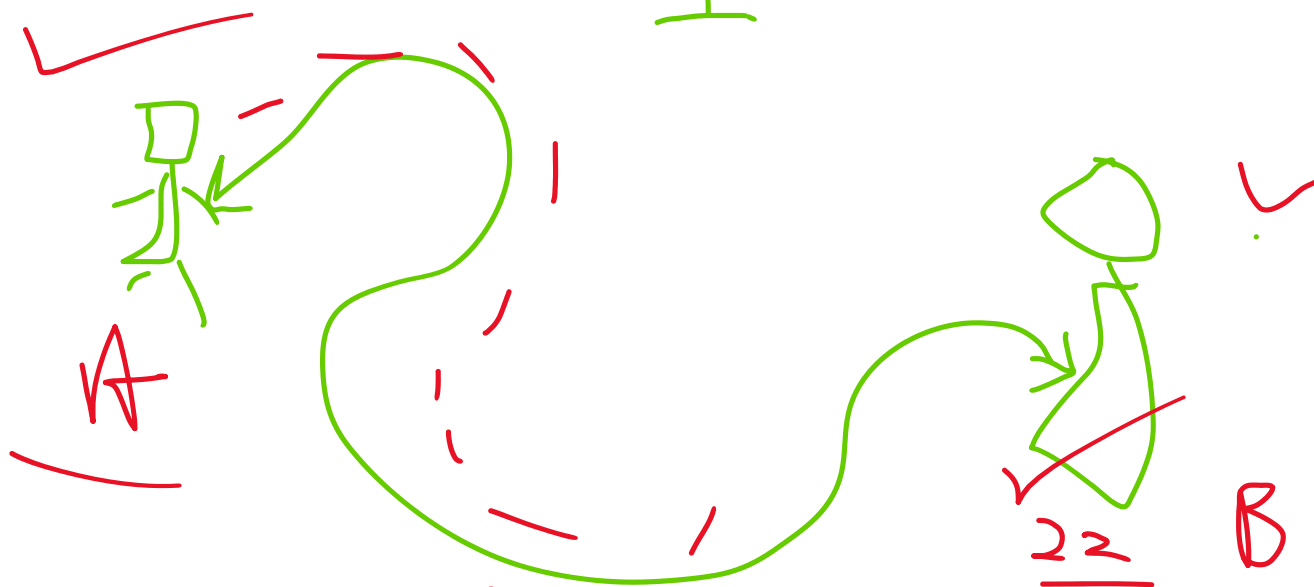


- The hacker above sends an ACK segment/packet on the first interaction (without three-way handshake).
 - If server returns no response means that might have a stateful firewall handling proper sessions
 - If server returns RST packet means that have no stateful firewall

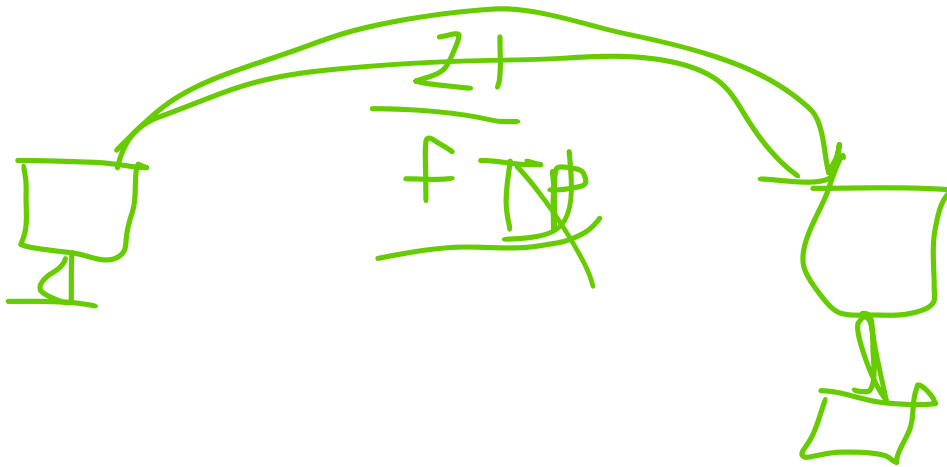
Fin

Ack + Fin

Ack



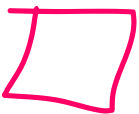
22

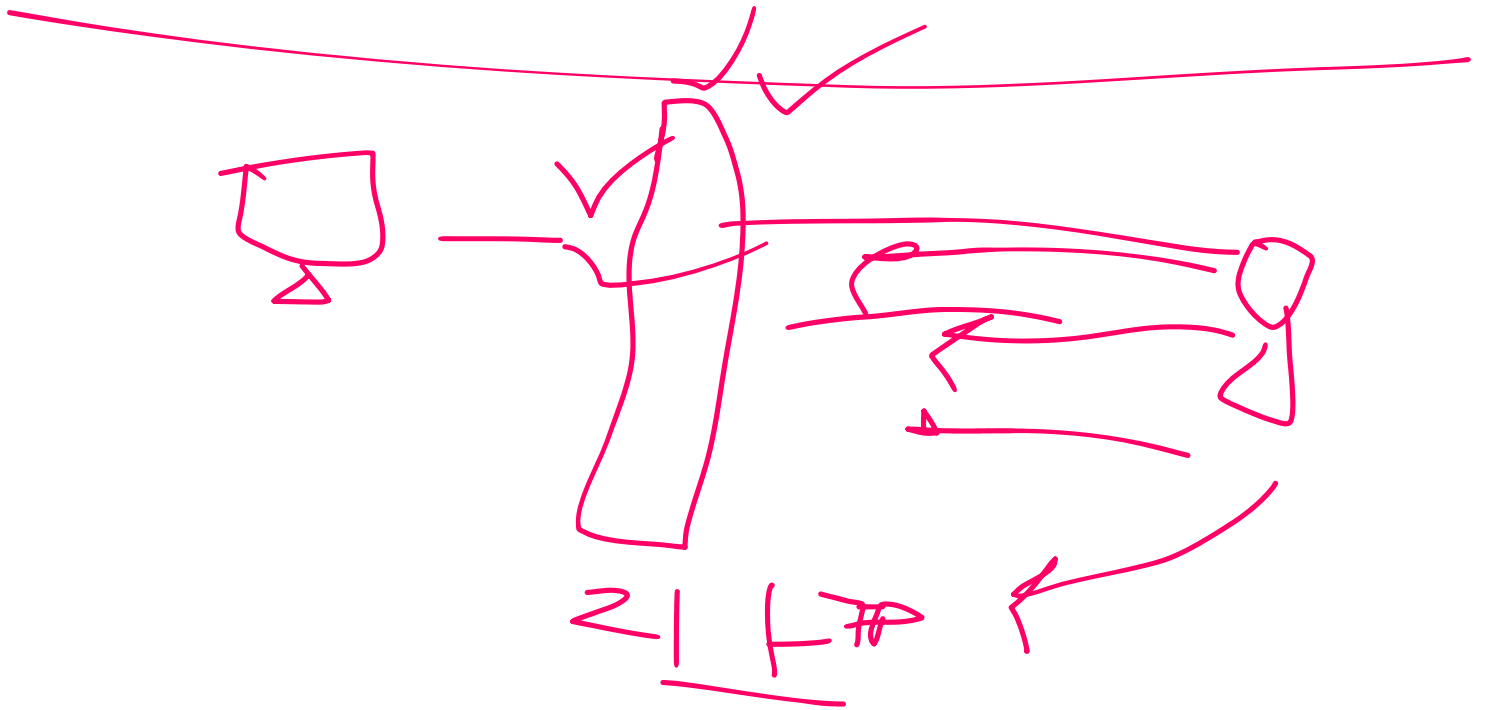
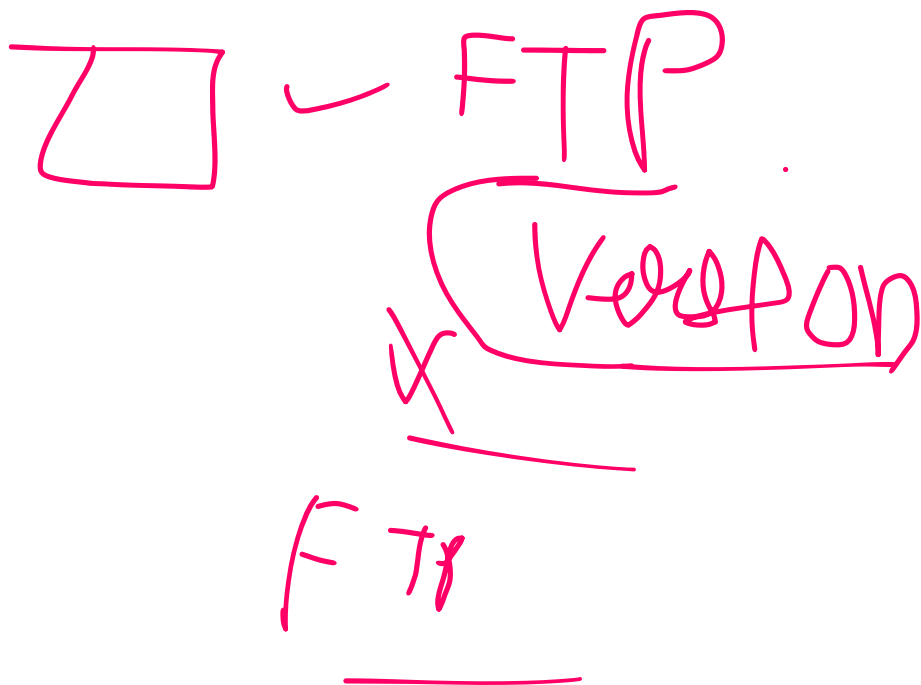


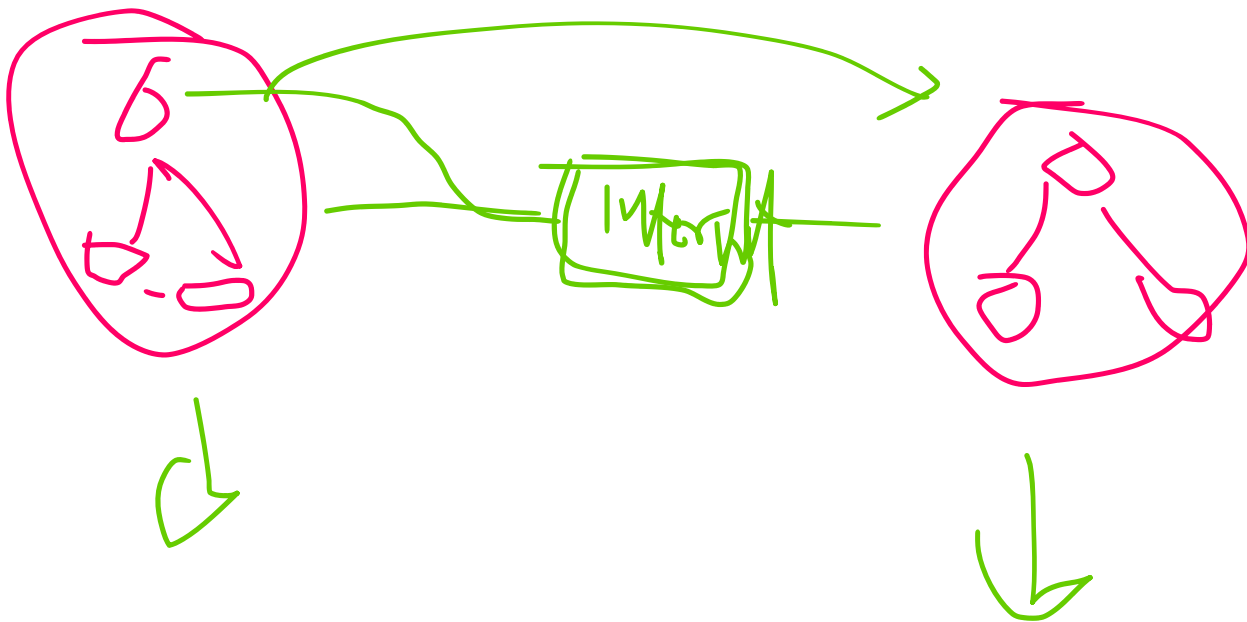
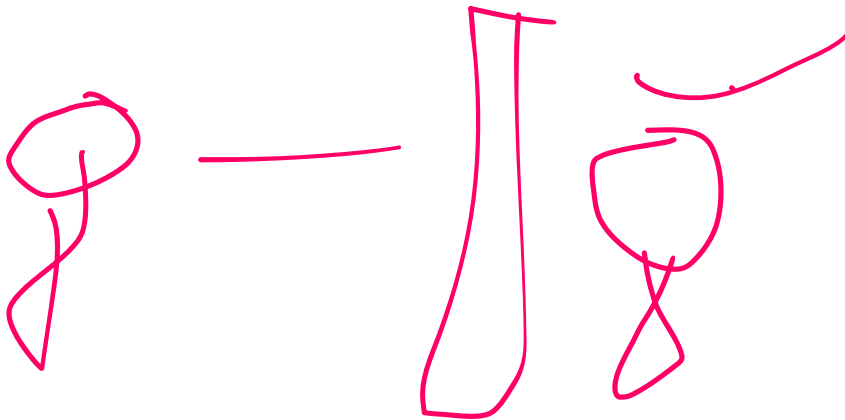
Well-Known Port Numbers

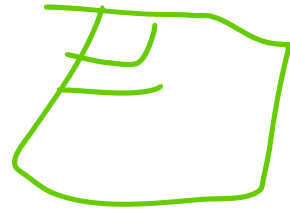
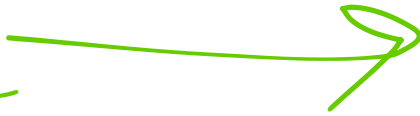
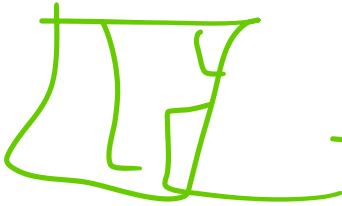
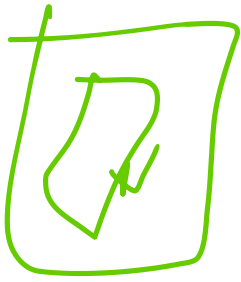


Service, Protocol, or Application	Port Number	TCP or UDP
✓ FTP (File Transfer Protocol)	20, 21	TCP
✓ SSH (Secure Shell Protocol)	22	TCP
- ✓ Telnet	23	TCP
SMTP (Simple Mail Transfer Protocol)	25	TCP
DNS (Domain Name System)	53	UDP
TFTP	68	UDP
HTTP	80	TCP
POP3	110	TCP
IMAP4	143	TCP
HTTPS	443	TCP









Natür

System

Web

Pyh

foergeri

GTF

141
GTF

17

CPH
H

OSCP → E4 like
Ofcp SKIM

